Tactix: Winner-Used PB Proof-Rule Gap

Report to Andrew

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Outcome

An exact replay of a retained SC24–26 LymphoSAT result establishes that a competition-winner solver actually used general pseudo-Boolean proof operations outside Tactix's current action-level proof surface. The selected internal solver was `tseitin-formulas`; its checked proof uses `red` extensions, local subproofs, `pol` arithmetic (including division), and `rup`. This upgrades the earlier AM402 source audit from an active-path possibility to a byte-identical baseline certificate witness. It does not identify a missing clausal RUP/RAT/SR primitive or prove that this formula cannot also be certified by DSR.

Verification and provenance

Baseline task 1659 recorded an 80,981-byte verified PBP for `tseitin_n188_d3`, but deleted the artifact and did not retain the internal producer label. AM408 job `1721206` replayed the cached original input through the same pinned LymphoSAT wrapper on hwulab (two CPUs, 8 GiB). The job completed `0:0`. The generated PBP has exactly the baseline SHA-256 `dd557484...22058bf1fd`; pinned VeriPB verified and elaborated it; pinned CakePB verified the elaboration, whose SHA-256 likewise matches the baseline. The baseline lists this UNSAT row as `SAT-only` for VBS eligibility; no VBS-score contribution is claimed. The exact script, input, full proofs, logs and hashes are frozen in the AM408 probe under `benchmarks/probes/`; the AM408 technical report in `benchmarks/reports/` gives the full hashes and limits.

Architecture implication and remaining work

Tactix currently emits PB arithmetic only for complete Lucky 19/20 startup refutations. There is no general registered live PB derivation action or proof-event interface after that boundary. Supporting this winner technique generally would require a typed PB surface and deliberate proof-format/state contract; embedding family-specific certificate code would not meet the project's shared-rule requirement. Native DSR remains canonical for search. The H2 no-option checker issue and post-finalization public-API model bug remain open, as do the Opus-owned AM400/401 fuzz campaigns and full Debug checks. N1–N4 and Rank/B+C/WV acceptance have not been earned, so the requested full Tier-D 5,000-second run has not been submitted.